

The integration of negative affect, pain and cognitive control in the cingulate cortex

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Abstract | It has been argued that emotion, pain and cognitive control are functionally segregated in distinct subdivisions of the cingulate cortex. However, recent observations encourage a fundamentally different view. Imaging studies demonstrate that negative affect, pain and cognitive control activate an overlapping region of the dorsal cingulate — the anterior midcingulate cortex (aMCC). Anatomical studies reveal that the aMCC constitutes a hub where information about reinforcers can be linked to motor centres responsible for expressing affect and executing goal-directed behaviour. Computational modelling and other kinds of evidence suggest that this intimacy reflects control processes that are common to all three domains. These observations compel a reconsideration of the dorsal cingulate's contribution to negative affect and pain.

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doi:10.1038/nrn2994

In humans and other primates, the cingulate — a thick belt of cortex encircling the corpus callosum — is one of the most prominent features on the mesial surface of the brain (FIG. 1a). Early research suggested that the rostral cingulate cortex (Brodmann's 'precingulate'; architectonic areas 24, 25, 32 and 33) plays a key part in affect and motivation² (FIG. 1b). More recent research has enlarged the breadth of functions ascribed to this region; in addition to emotion³, the rostral cingulate cortex has a central role in contemporary models of pain^{4,5} and cognitive control^{6,7}. Work in these three basic domains has, in turn, strongly influenced prominent models of social behaviour⁸, psychopathology^{9–11} and neurological disorders¹².

Despite this progress, key questions about the functional organization and significance of activity in the rostral cingulate cortex remain unresolved. Perhaps the most basic question is whether emotion, pain and cognitive control are segregated into distinct subdivisions of the rostral cingulate or are instead integrated in a common region. In a pair of landmark reviews, Devinsky *et al.*¹³ and Bush *et al.*¹⁴ marshalled a broad range of functional imaging, electrophysiological and anatomical data in support of functional segregation, arguing that the anterior cingulate cortex (ACC; also known as the 'rostral' ACC) is specialized for affective processes, whereas the midcingulate cortex (MCC; also known as the 'dorsal' ACC) is specialized for cognitive

processes (FIG. 1c,d). Subsequent meta-analyses of functional imaging studies have provided some support for this claim¹⁵.

Although the segregationist model remains highly influential, new data suggests that it is no longer tenable. For example, recent imaging data implicate MCC in the regulation of autonomic activity^{16,17} and the perception and production of emotion^{3,18}. Similarly, neuronal recordings demonstrate that MCC is responsive to emotionally charged words in humans¹⁹. Especially robust links have been forged between activity in the anterior subdivision of the MCC (aMCC; FIG. 1c) and the experience of more intense states of negative affect, as with the anticipation^{20–22} and delivery^{23,24} of pain and other kinds of aversive stimuli^{25,26}. A particularly dramatic example comes from a recent study showing that activation of aMCC parametrically tracks the physical imminence of a spider placed near the foot²⁷. Importantly, meta-analyses that have examined imaging studies of negative affect²¹, pain²³ or cognitive control²⁸ in isolation suggest that each of these domains consistently activate aMCC. Based on such observations, there is a growing recognition that aMCC might implement a domain-general process that is integral to negative affect, pain and cognitive control^{15,29–34}.

In this Review, we examine this integrative hypothesis about the functional organization of the rostral cingulate cortex with a special focus on the contribution of

Cognitive control

A range of elementary processes (such as attention, inhibition and learning) that are engaged when automatic or habitual responses are insufficient to sustain goal-directed behaviour. Control can be engaged proactively or reactively.

aMCC to negative affect and pain. We neither attempt a comprehensive overview (see REF. 12) nor do we provide a detailed discussion of this region's role in appetitively motivated learning and behaviour, phenomena that have been the subject of other recent reviews^{35–37}. We first address the question of whether MCC should be conceptualized as a territory specialized for 'cognitive' processes, as segregationist models claim. We show how three largely independent lines of evidence — physiological, anatomical and functional — challenge

longstanding claims of functional segregation in the rostral cingulate cortex. We then explore the possibility of using ideas adopted from computational models of cognitive control and reinforcement learning to address the contribution of aMCC to negative affect and pain. Although these models are familiar to many cognitive neuroscientists, we believe that they provide a useful, if underappreciated, framework for generating mechanistic hypotheses about the role of aMCC in aversively motivated behaviour. This perspective, which we term the 'adaptive control hypothesis', can account for a number of observations not readily accommodated by segregationist models. However, it also raises a number of interesting new questions. We conclude by outlining several strategies for answering them.

Anatomical and physiological convergence

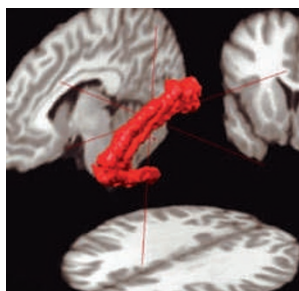
Functional imaging evidence of overlap in the aMCC.

As the size and scope of the imaging literature have burgeoned, it has become increasingly difficult to synthesize new data into existing models of functional organization. This problem is particularly acute when attempting to integrate observations from disparate domains, such as affect, pain and cognition. This challenge can be overcome using new techniques for performing voxel-by-voxel, or 'coordinate-based', meta-analysis (CBMA)³⁸. Here we used CBMA to evaluate whether imaging studies of negative affect, pain and cognitive control provide evidence for colocalization or segregation in the rostral cingulate cortex. Given the observations described earlier, we anticipated that all three domains would consistently activate an overlapping region within aMCC. To do so in an unbiased and replicable way, we identified 939 studies in the [BrainMap database](#) reporting activation in ACC or MCC. We then identified activation foci (peaks) associated with manipulations of negative affect, pain or cognitive control in healthy unmedicated adults (for additional details, see Supplementary information S1 (box)).

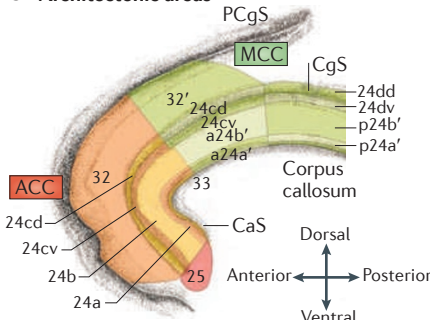
The negative affect database included foci associated with manipulations designed to induce negative emotions, including fear, anger and disgust. To minimize potential overlap with studies of cognition, manipulations that were unlikely to produce clear-cut affect — such as the perception of facial expressions or the reading of 'taboo' words — were excluded. The pain database included foci associated with the delivery of physically painful stimuli, such as heat, cold or electric shock. The cognitive control database included foci associated with a number of tasks designed to isolate the need to overcome the reflexive allocation of attention or execution of actions (for example, the Stroop task, Go/No-Go task and Eriksen Flanker task). Collectively, the three databases included 380 activation foci from 192 studies involving nearly 3,000 participants (FIG. 2).

We used the activation likelihood estimate (ALE) algorithm³⁸ to identify voxels within ACC or MCC that were consistently activated by negative affect, pain or cognitive control (see Supplementary information S1 (box)). Using these three maps, we created a single 'conjunction map'³⁹ showing voxels that were consistently activated across the three domains. If negative

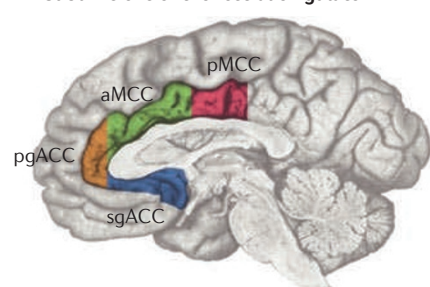
a Left rostral cingulate cortex



b Architectonic areas



c Subdivisions of the rostral cingulate



d Functional segregation model

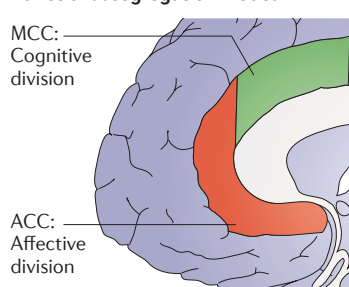
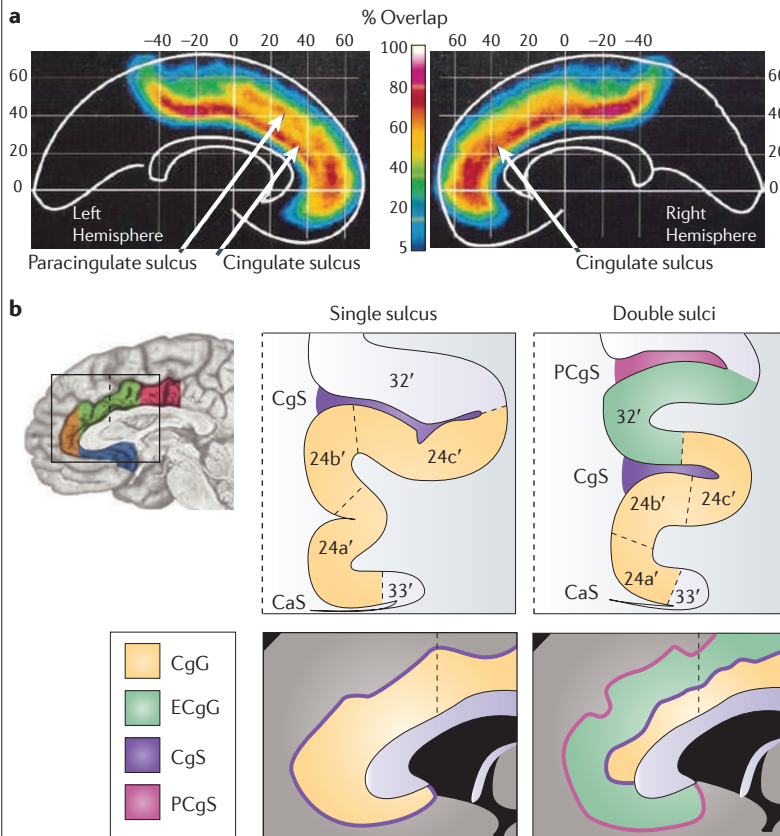


Figure 1 | Divisions of the human rostral cingulate cortex. The rostral cingulate has been partitioned on physiological and anatomical grounds at spatial scales ranging from the macroscopic to the molecular. **a** | Three-dimensional rendering of the left rostral cingulate cortex. The cingulate (shown in red) was manually traced on a single subject's magnetic resonance image (MRI). Much of the constituent cortical grey matter lies buried within the cingulate sulci, a fact not apparent from inspection of the mesial surface (BOX 1; see [Supplementary information S1 \(box\)](#)). **b** | Architectonic areas of the cingulate. Areas were defined²¹¹ on the basis of differences in microanatomy and neurotransmitter chemistry, and hence, differ somewhat from the classical descriptions of Brodmann and other pioneering neuroanatomists^{1,212}. Architectonic features provide one means of defining homologies across species^{183,213}. **c** | The four major subdivisions of the rostral cingulate. Subdivisions were defined by Vogt and colleagues²¹³ on the basis of regional differences in microanatomy, connectivity and physiology. The supracallosal portion of the cingulate is designated the midcingulate cortex (MCC) and is divided into anterior (aMCC; shown in green) and posterior (pMCC; shown in magenta) subdivisions. The portion of the cingulate lying anterior and ventral to the corpus callosum is designated the anterior cingulate cortex (ACC) and is divided into pregenual (pgACC; shown in orange) and subgenual (sgACC; shown in cyan) subdivisions by the coronal plane at the anterior tip of the genu (see also [Supplementary information S1 \(box\)](#)). **d** | The functional segregation model of Bush, Luu and Posner¹⁴. On physiological and anatomical grounds, Bush *et al.*¹⁴ proposed that the rostral cingulate consists of two functionally segregated regions: a rostroventral 'affective' division (ACC; originally termed ventral ACC) and a dorsal 'cognitive' division (MCC; originally termed dorsal ACC). PCgS, paracingulate sulcus. CaS, callosal sulcus; CgS, cingulate sulcus. Part **b** is reproduced, with permission, from REF. 211 © (2009) John Wiley & Sons. Part **c** is reproduced, with permission, from REF. 198 © (1989) Oxford University Press; Part **d** is modified, with permission, from REF. 14 © (2000) Cell Press.

affect, pain and cognitive control are strictly segregated, we would expect to see little or no overlap. Instead, the conjunction map revealed a sizable cluster in the dorsal portion of the aMCC (FIG. 2).

Box 1 | Individual differences in rostral cingulate anatomy



Individual differences in the macroscopic anatomy of the cingulate represent a key obstacle to resolving the finer details of this region's functional organization. In particular, there is considerable variability in the paracingulate sulcus (PCgS), a tertiary sulcus that is present in about one-half of the population and more prominent in the left hemisphere (see the figure, part a)^{190,191}.

The presence of this sulcus exerts a strong impact on the layout and relative volume of the architectonic areas comprising MCC (see the figure, part b). In particular, area 32', which is otherwise found in the depths of the cingulate sulcus (CgS), expands to occupy the crown of the external cingulate gyrus (ECgG; the 'superior' or 'paracingulate' gyrus)¹⁹². A parallel reduction occurs in the size of the more ventral supracallosal areas occupying the cingulate gyrus (CgG; areas 24a' and 24b')^{191,192}. A key consequence is that the size and spatially normalized location of the cingulate premotor areas harboured within MCC (areas 32' and 24c'; FIG. 3) can vary substantially across individuals.

More generally, variation in sulcal anatomy will tend to obscure fine-grained distinctions between deep and superficial strata within each of the major subdivisions; that is, unmodelled variation in the cingulate sulci will tend to inflate the spread of activation clusters and hamper efforts to dissociate superior from inferior areas within MCC (REFS 193, 194) and rostral from caudal areas within ACC (compare with FIGS 1, 3). Accounting for such individual differences may permit a clearer separation of intermingled affective, nociceptive and cognitive processes within aMCC (as in several important early imaging studies of pain^{195,196}). CaS, callosal sulcus. Part a is reproduced, with permission, from REF. 197 © (1996) Oxford University Press. Numbers along the vertical and horizontal axes indicate the distance (mm) from the vertical and horizontal planes, respectively, defined by the anterior commissure. Part b, top left panel is modified, with permission, from REF. 198 © (1989) Oxford University Press; Part b, bottom panels are modified, with permission, from REF. 192 © (1995) John Wiley & Sons.

Another way to evaluate functional segregation is to test whether each domain differentially activates the 'cognitive' division (MCC) compared to the 'affective' division (ACC) of the cingulate cortex (FIG. 1d). In the case of strict segregation, we would expect studies of cognitive control to activate MCC more frequently than ACC, and indeed, this is what was found (odds = 4.8, confidence interval (CI) = 3.2–7.1, $P < 0.001$). Conversely, we would expect studies of negative affect to activate MCC less frequently than ACC, but in fact they were equally likely to activate the two divisions (odds = 1.1, CI = 0.8–1.6, $P = 0.64$). It is less clear what to expect for studies of pain, but given the strong association between pain and negative affect⁴⁰, we might expect pain to preferentially activate the 'affective' division (ACC). Instead, studies of pain were more likely to activate the 'cognitive' division (MCC) (odds = 4.9, CI = 2.9–8.3, $P < 0.001$).

Collectively, these observations refute claims that cognition and emotion are strictly segregated into different divisions of the rostral cingulate cortex — claims that were heavily based on an early meta-analysis of imaging studies¹⁴ (for a discussion of why our results differed from earlier analyses, see Supplementary information S1 (box)). Instead, these observations show that aMCC is consistently activated by the elicitation of negative affect, pain and cognitive control. Of course, these results do not preclude the possibility that this region contributes to other psychological processes, such as reward-motivated behaviour. Furthermore, they do not address whether segregation is present at finer levels of analysis — for example, in individual participants or neurons. Similarly, segregation may be present on a finer timescale than that resolved by conventional imaging techniques³⁰. Nevertheless, what these results do demonstrate is that conventional functional imaging studies of negative affect, pain and cognitive control all consistently report activation in this subdivision of rostral cingulate cortex.

Anatomical evidence of integration. It has often been suggested that the MCC possesses few connections with regions of the brain implicated in affect, motivation and nociception¹⁴. However, several recent tracing studies, along with a few older ones, indicate that this is not the case. In the remainder of this section, we focus largely on invasive tracing studies performed in monkeys — although rapid progress has been made in refining techniques for mapping structural connectivity in the living human brain, invasive studies are still considered the gold standard⁴¹. These data suggest that aMCC represents a hub, where information about pain, and other, more abstract kinds of punishment and negative feedback could be linked to motor centres responsible for expressing emotion on the face and coordinating aversively motivated instrumental behaviours.

The aMCC harbours the rostral cingulate zone (RCZ), a somatotopically organized premotor area⁴². Originally identified on the basis of physiological and anatomical criteria in the monkey (in which it is termed the rostral cingulate motor area), the RCZ has been provisionally identified in humans with Brodmann areas 32' and 24c' in the vicinity of the cingulate sulcus^{43,44} (BOX 1; FIGS 1b, 3a).

The RCZ projects to the spinal cord, dorsal (sensorimotor) striatum and primary motor, premotor and supplementary motor cortices. Physiological studies in humans and monkeys indicate that RCZ is sensitive to the more abstract aspects of action planning and inhibition^{42,45}. This

stands in contrast to the caudal cingulate zone (CCZ), lying at the junction of aMCC and pMCC (FIG. 3a,b), which has been linked to more specific motor parameters, such as the precise direction of movement^{42,45}.

Other data suggest that RCZ can contribute to the expression of affect on the face (FIG. 3c). In monkeys, RCZ sends heavy bilateral projections to neurons in the facial nucleus that, in turn, innervate the muscles of the upper face (for example, the corrugator, frontalis and orbicularis oculi muscles)⁴⁶, the same muscles that underlie the expression of emotion and pain in monkeys⁴⁷ and humans^{48,49} (FIG. 3d). Indeed, direct microstimulation of RCZ in monkeys can evoke facial displays classically associated with the fight-or-flight reaction⁴⁷. However, the precise role of RCZ in the wilful or spontaneous expression of emotion, or the regulation of such expressions remains unknown.

For the remainder of this Review, we refer to the cluster of activation overlap obtained in our meta-analysis as aMCC (FIG. 2). Nevertheless, the relatively dorsal position of the cluster within aMCC (approximately corresponding to architectonic areas 32' and a24b'/c'; FIGS 1b,2) is consistent with the provisional location of RCZ⁴⁴. This suggests that it is specifically RCZ that is commonly activated by imaging studies of negative affect, pain and cognitive control.

The aMCC is also characterized by substantial connections with subcortical regions involved in negative affect and pain (FIG. 4). It is a primary cortical target of the spinothalamic system, the chief source of peripheral nociceptive information⁵⁰. There is some evidence that it sends connections to the lateral column of the periaqueductal gray (IPAG), a region that is closely linked to vigilance, fight-or-flight and other defensive responses in rats and cats⁵¹. Robust reciprocal connections have also been found between aMCC and the lateral basal nucleus of the amygdala^{52,53}. Functional connectivity data from humans show a similar pattern^{54,55}. The basal nucleus is a convergence zone for information from the lateral nucleus of the amygdala (crucial for the initial evaluation of motivationally significant stimuli) and orbitofrontal cortex (OFC; a key source of inhibitory inputs to the amygdala)⁵⁶. In rodents, the basal nucleus has also been implicated in the learning of aversively motivated instrumental behaviours⁵⁷.

The aMCC projects to the ventral striatum, including the core region of the nucleus accumbens⁵⁸. Although the ventral striatum is commonly associated with reward and appetitively motivated behaviour, it is also activated by the anticipation and avoidance of pain^{59,60} and other aversive stimuli in humans^{59,61,62}. Dopaminergic inputs to aMCC in the monkey are predominantly from the substantia nigra and retrorubral area, with a weaker contribution from the ventral tegmental area⁶³. Interestingly, some neurons in the primate substantia nigra are activated by aversive stimuli and cues predicting their occurrence⁶⁴, suggesting that information about reinforcers, including punishment, could be passed to aMCC via ascending dopaminergic pathways.

In the cortex, aMCC is reciprocally connected with frontoparietal regions implicated in cognitive control

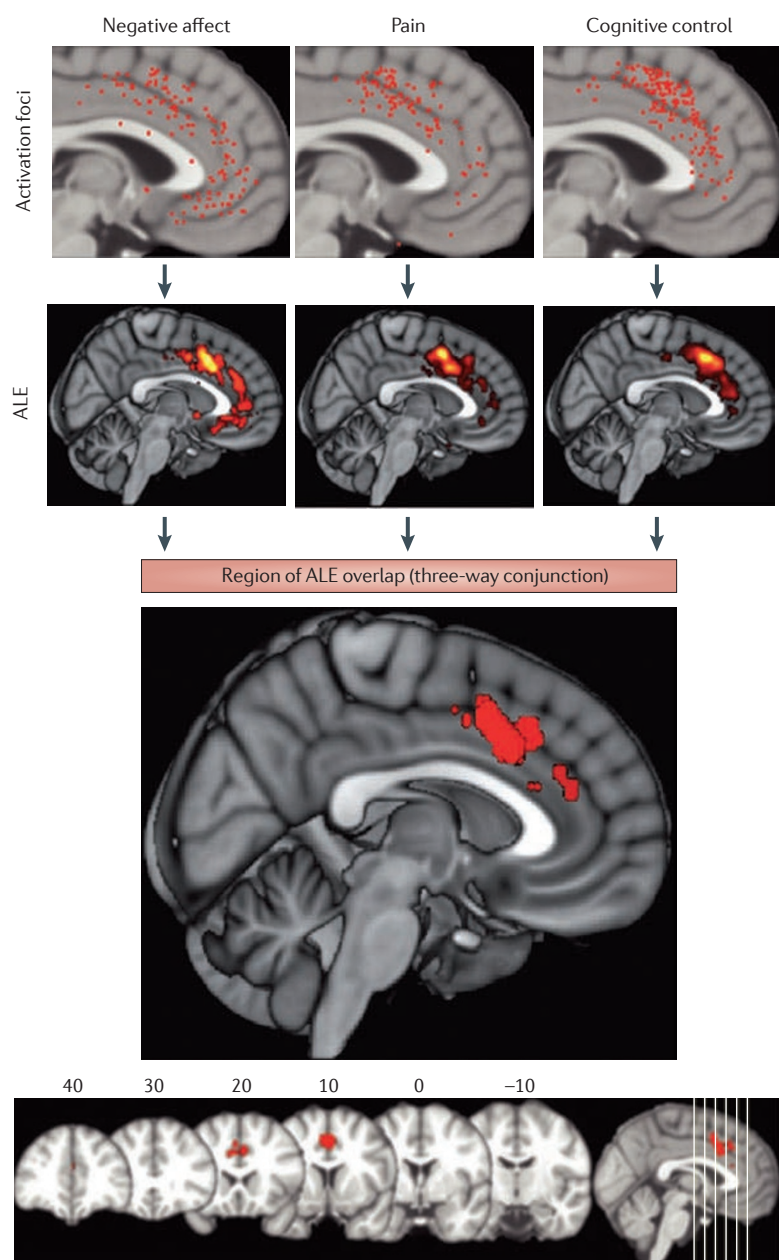


Figure 2 | Negative affect, pain and cognitive control activate a common region within the aMCC. The map depicts the results of a coordinate-based meta-analysis (CBMA) of 380 activation foci derived from 192 experiments and involving more than 3,000 participants. The uppermost panel shows the spatially normalized foci for each domain. The next panel shows thresholded activation likelihood estimate (ALE)^{38,214} maps for each domain considered in isolation. The two lowest panels depict the region of overlap across the three domains. The red cluster indicates the location of a three-way minimum significance conjunction³⁹ of the three domains. The cluster lies in the anterior midcingulate cortex (aMCC) in the vicinity of areas 32' and a24b'/c' (Talairach coordinates: $x = 0$, $y = 12$, $z = 42$; volume is 11680 mm³). No other cluster reached significance. The numbers indicate distance (in mm) from the anterior commissure (for additional methodological details and results, see Supplementary information S1 (box)).

Computational model

A mathematically detailed simulation of a psychological construct that can afford quantitative predictions of trial-by-trial fluctuations in behaviour and neurophysiology.

Reinforcement learning models

(Often abbreviated to RL models.) A class of computational models describing how organisms learn to maximize reinforcement based on experience. RL models assume that organisms update reinforcer expectations on the basis of prediction errors and the current learning rate.

Stroop task

A task in which subjects rapidly respond to a colour word, such as 'blue', on the basis of the colour in which the letters are displayed. The task is easy when the colour and word are compatible ('blue' depicted in blue), but is more difficult when the two are incompatible ('blue' depicted in red).

Go/No-Go task

A task in which subjects must rapidly respond to one kind of cue ('Go') while withholding responses to another ('No-Go').

Eriksen Flanker task

A task in which subjects rapidly respond to a centrally presented visual cue, such as an arrowhead, that is neighboured (flanked) by cues that can potentially code an alternative response.

Instrumental behaviour

Behaviour that is goal-directed insofar as it increases the likelihood of obtaining rewards or avoiding punishments. Instrumental behaviour is distinguished from behaviours that are reflexively elicited independent of reinforcement, as in Pavlovian (classical) conditioning.

Reinforcer

A stimulus that is capable (intrinsically or through learning) of eliciting instrumental behaviour; reward and punishment.

Attentional set

A template, rule or goal held in memory to guide attention (for example, search for angry faces in a crowded visual scene).

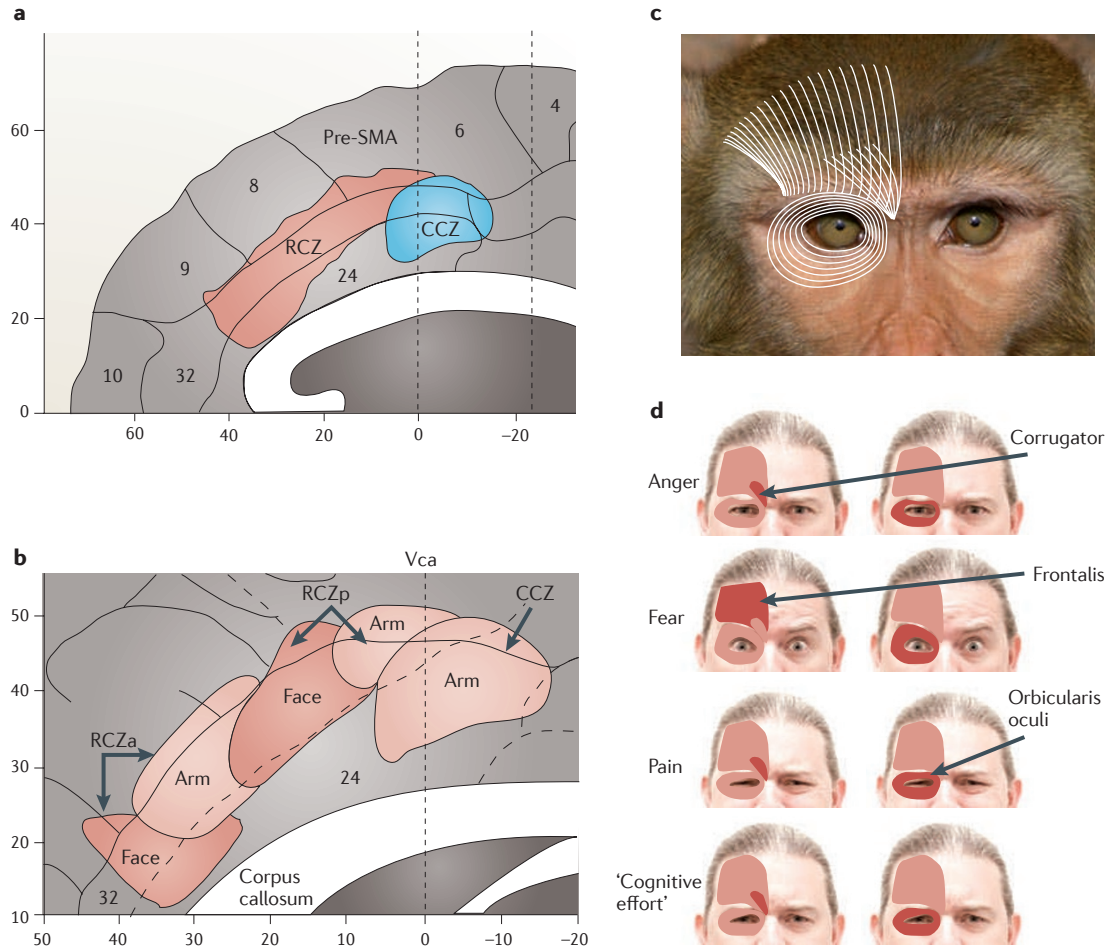


Figure 3 | Cingulate premotor areas in the human MCC. **a** | Provisional locations of the rostral and caudal cingulate zones (RCZ and CCZ)^{6,43}. The RCZ lies in the anterior midcingulate cortex (aMCC), whereas the CCZ lies at the junction of aMCC and posterior MCC (pMCC) (FIG. 1). Zone borders are approximations (see also REF. 44). **b** | Somatotopy in RCZ and CCZ based on human imaging studies⁴³. **c** | Combined tracing and microstimulation work in macaque monkeys indicates that the monkey analogue of the human RCZ projects to the facial nucleus^{50,215}, allowing it to control the muscles of the upper face (shown in white for the macaque). The facial muscles are largely conserved across primate species^{216,217}. **d** | In humans, the muscles of the upper face (shown in red) have been associated with the elicitation of negative affect (for example, anger and fear), pain and — consistent with Darwin's suggestions²¹⁸ — perhaps 'cognitive effort' as well (see also Supplementary information S1 (box)). Pre-SMA, pre-supplementary motor area; RCZa, anterior portion of the RCZ; RCZp, posterior portion of the RCZ; Vca, anterior commissure. Numbers along the vertical and horizontal axes indicate the distance (mm) from the vertical and horizontal planes, respectively, defined by the anterior commissure. Image in part **c** courtesy of B. Waller, University of Portsmouth, UK, and L. Parr, Emory University, USA. Images in part **d** courtesy of J. Coan, University of Virginia, USA and C. Thrasher. Part **a** is modified, with permission, from REF. 6 © (2004) AAAS. Part **b** is modified, with permission, from REF. 43 © (1996) Oxford University Press.

and the maintenance of goals (such as attentional sets and rules), including dorsolateral prefrontal cortex (architectonic area 9/46)^{65,66}. However, it is also connected with all major divisions of the insula^{67,68}, a region strongly implicated in affect⁶⁹, pain^{70,71} and cognitive control (including responses to errors and negative feedback)^{72–75}.

Collectively, these data show that aMCC is well positioned to synthesize information about unlearned reinforcers (for example, pain, predators and threatening conspecifics) and learned reinforcers (for example, aversive cues and negative feedback) with current goals. Through efferents targeting the facial nucleus, aMCC could exploit this blend of information to drive or, more

likely, to flexibly regulate⁷⁶ the expressions needed to visually communicate with conspecifics and potential predators at close range. Such signals are a key element of many species' defensive repertoire⁷⁷, including that of our closest living relative, the chimpanzee⁷⁸. The value of such expressions is not limited to communication; other evidence suggests that they serve to optimize perception, amplifying or attenuating the intake of sensory information⁷⁹. Finally, the abundant connections linking aMCC to other motor centres would permit it to use information about reinforcers to plan or refine more complex, aversively motivated instrumental behaviours. This stands in sharp contrast with other cortical

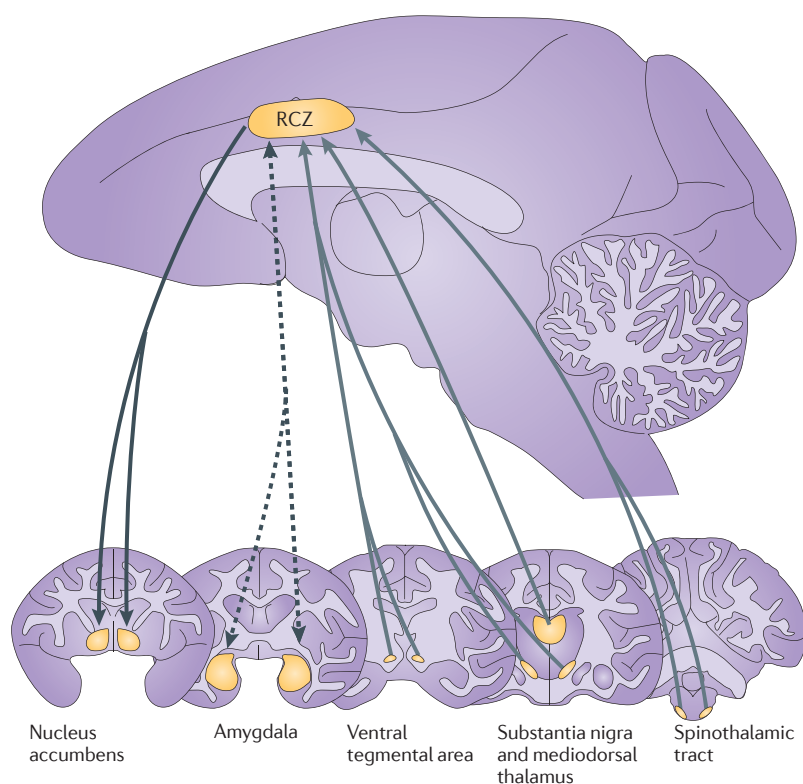


Figure 4 | Subcortical connectivity of the macaque analogue to the human RCZ. The monkey analogue to the rostral cingulate zone (RCZ) receives substantial inputs from the spinothalamic system, which relays nociceptive information from the periphery to RCZ via the mediodorsal nucleus of the thalamus. Dopaminergic inputs to RCZ arise from the substantia nigra and, to a lesser extent, the ventral tegmental area. RCZ projects to the ventral striatum, including the core region of nucleus accumbens, and has robust reciprocal connections with the lateral basal nucleus of the amygdala (see [Supplementary information S1](#) (box)). Afferents are shown in black, efferents are shown in light grey and reciprocal connections are shown by dotted arrows.

regions implicated in affect and motivation, such as the OFC and insula, which lack strong ties with motor centres^{35,80}.

Evidence of functional convergence. Our meta-analysis revealed that negative affect, pain and cognitive control consistently activate an overlapping region of aMCC. This overlap suggests the possibility that aMCC performs a similar role across domains (for additional discussion of the logic underlying this inference, see [Supplementary information S1](#) (box)). The anatomical data reviewed in the previous section are consistent with this hypothesis. We next consider whether the three domains also exhibit convergent functional properties. The logic here is that if aMCC implements a single, domain-general function, then measures of negative affect, pain and cognitive control should covary. These measures should also respond similarly to particular experimental manipulations and covary with distinct individual differences.

Several lines of evidence indicate that negative affect, pain and cognitive control exhibit a measure of functional convergence. First, individual differences in measures of MCC structure predict variation in trait negative affect (neuroticism)⁸¹, conditioned fear⁸² and cognitive

control⁸³. Broadly speaking, individuals with a larger MCC report that they are predisposed to experience greater negative affect, exhibit enhanced electrodermal activity (EDA) and neural activation in aMCC during aversive conditioning tasks, and show reduced interference when performing the Stroop task⁸⁴. Moreover, individual differences in negative affect predict variation in the other two domains. Specifically, individuals characterized by greater negative affect show increased engagement of control processes (indexed by well-validated event-related potential (ERP) measures⁸⁵ that are thought to be generated in MCC⁸⁶) when performing prototypical cognitive control tasks (see [Supplementary information S1](#) (box)). They also exhibit increased sensitivity to experimental pain, particularly the affective qualities of pain (pain ‘unpleasantness’)^{87–90}.

Second, manipulations of all three domains have been shown to amplify measures of autonomic arousal and negative affect. In particular, pain^{91,92} and cognitive control^{93,94} have been shown to increase EDA and amplify the fear-potentiated startle reflex ([FIG. 3d](#)). These findings are linked to MCC by the observation that individuals who exhibit larger startle reflexes in response to errors on a prototypical cognitive control task (the Eriksen flanker task) show ERP evidence of enhanced control-related activity in MCC⁹³. Similarly, individuals showing increased EDA in response to pain exhibit greater activation in aMCC and amygdala⁹¹.

Third, manipulations of all three domains can produce distinct changes in the muscles of the upper face^{48,49,95–98} ([FIG. 3d](#)). As noted earlier, tracing studies in monkeys suggest that these muscles can be modulated, through the facial nucleus, by aMCC.

Fourth, manipulations targeting one domain can alter measures of the others. Experimentally induced negative affect, for example, can selectively disrupt the performance of tasks that strongly engage cognitive control^{48,99}. Cognitive control tasks can attenuate the intensity of negative affect¹⁰⁰ and pain^{101,102}. Indeed, concurrent performance of the Stroop task attenuates pain-evoked activation in MCC¹⁰³. Analgesic placebos show evidence of ‘cross-domain transfer’ — that is, they attenuate negative affect elicited by aversive images in addition to decreasing pain¹⁰⁴. Conversely, the administration of anxiolytic compounds (that are not directly analgesic) can reduce the experience of pain⁸⁹ and reduce aMCC activation to cues predictive of imminent pain¹⁰⁴. Evidence for cross-domain interactions is consistent with the idea that negative affect, pain and cognitive control can compete for, or otherwise modulate, a common functional resource implemented in aMCC. Cross-domain disruption, in particular, indicates that this resource makes a necessary contribution across domains. It is important to emphasize, however, that such cross-domain influences are often complex and do not necessarily impair performance or attenuate the intensity of subjective experience^{48,102}.

Fifth, all three domains are similarly affected by manipulations of ‘certainty’, variously described in terms of ambiguity (‘unknown uncertainty’ of an outcome), controllability, determinacy, predictability, risk (‘known

Architectonic area

A region of the brain defined by its cellular and molecular neuroanatomy, including neuronal structure (cytoarchitecture), myelin structure (myeloarchitecture) and neurochemistry (chemoarchitecture).

Electrodermal activity

(Often abbreviated to EDA.) Changes in the electrical resistance of the dermis stemming from activity of the sweat glands. EDA reflects activation in the sympathetic nervous system and is used to index arousal, stress and cognitive load.

Box 2 | Mapping neurobiological models of control onto the aMCC

Control is thought to reflect two elementary processes — one responsible for monitoring performance and detecting the need for control (a monitor), the other responsible for implementing control to protect and optimize goal-directed behaviour (a controller) — that together form a closed feedback loop. Control processes are often conceptualized as top-down signals that bias competition among stimuli (for attention) or response options (for action).

Some of the most fundamental computational and neurophysiological details of the rostral cingulate cortex's contribution to control remain contentious^{7,146,199–202}. In particular, a number of proposals have been made about what is monitored, including errors, error likelihood, expected risk, response conflict and reinforcement volatility^{199,203}. Likewise, the control process has been modelled as a variety of different biasing signals, including biases toward slower (that is, more cautious) action, increased focusing of attention (that is, increasing the amount of attention allocated to relevant sensory information and/or decreasing the amount allocated to irrelevant or distracting information) or changes in the rate of new learning^{199,204}.

Although it is clear that the anterior midcingulate cortex (aMCC) plays a key part in control, it is not yet clear whether this region is best conceptualized as a monitor, responsible for triggering control processes implemented in other regions (for example, the lateral prefrontal cortex and the striatum) in response to a locally generated signal, such as response conflict^{20,205}; as a controller, triggered by signals conveyed from other regions, such as the striatum²⁰⁶; or some more complex arrangement^{73,146,200,205}. It may also be the case that different kinds of control are implemented in neighbouring subregions of aMCC and pregenual anterior cingulate cortex (pgACC) or, perhaps, are organized along a more continuous functional gradient^{18,44,160–162}. Similar ambiguities apply to scalp-recorded event-related potential (ERP) measures of control that are linked to MCC²⁰⁷. Research to clarify them is likely to have substantial benefits for understanding the part played by aMCC in negative affect and pain. Work that weds computational modelling, meta-analysis and individual differences analyses with neurophysiological techniques is likely to prove especially fruitful²⁰¹.

uncertainty' of an outcome) or volatility. For example, reducing the predictability of a physical threat amplifies ratings of anxiety and peripheral measures of negative affect, such as fear-potentiated startle and EDA^{105,106}, and activates aMCC¹⁰⁶. Likewise, uncertainty about the timing or magnitude of painful stimulation increases ratings of pain unpleasantness and can markedly alter the psychophysical function relating different 'doses' of painful stimulation to subjective perception^{105,107–110}. Reductions in perceived instrumental control have been shown to amplify pain-evoked activation in aMCC¹¹¹ and increase preparatory MCC activity in aversively motivated instrumental conditioning paradigms¹¹². Moreover, ERP indices of cognitive control that are thought to be generated in MCC are amplified by response uncertainty⁸⁶ and unexpected outcomes¹¹³. Along similar lines, greater response uncertainty during probabilistic learning¹¹⁴ and economic decision making tasks^{115,116} activate aMCC. Taken together, these observations suggest that the common function implemented in aMCC is sensitive to certainty about 'actions' (which response to make) and 'outcomes' (the magnitude and likelihood of the reinforcers acquired or avoided by such actions).

Finally, the hypothesis that aMCC activation could reflect a common operation across domains is consistent with striking similarities in the functions that have been ascribed to negative affect, pain and cognitive control by domain-specific theories. Cognitive control, for example, has been described as an early warning system that allows animals to proactively alter attention or

behaviour to avoid future errors¹¹⁷. A similar warning function has been ascribed to pain¹¹⁸ and to some kinds of negative affect, such as fear and anxiety^{119,120}. Like cognitive control, it has been suggested that negative affect (for example, fear and anger) is goal-directed and flexibly coordinates anticipatory responses that decrease the likelihood of future punishment^{121,122}. Demands for cognitive control are also thought to motivate new learning³⁴. Such demands may serve as a teaching signal that penalizes choices, strategies or actions requiring greater control, promoting avoidance of cognitively taxing actions in the future. Indeed, it has been shown that variation in the error-related negativity (ERN; an ERP component that is sensitive to control demands and thought to be generated in MCC) predicts the degree to which individuals learn from the negative consequences of their actions: individuals with larger ERNs show enhanced avoidance learning for events associated with negative outcomes^{123,124}. Imaging studies have revealed broadly similar effects in MCC¹²⁵. This teaching function is reminiscent of the role ascribed to pain and negative affect in reinforcing withdrawal-related behaviours and driving the acquisition of instrumental avoidance^{12,119,121}. Given these numerous parallels, Yeung and colleagues³⁰ have speculated that the signals responsible for triggering cognitive control (for example, the output of a response conflict monitor; BOX 2) could represent the computational underpinning of negative affect. That is, the same computational machinery might be engaged when cognitive control or negative affect are elicited.

The adaptive control hypothesis

The aMCC uses information about punishment to control aversively motivated actions. The observations reviewed in the previous section suggest that aMCC makes a similar functional contribution to negative affect, pain and cognitive control. But what is the nature of this 'domain-general' contribution? A plausible working hypothesis is that negative affect and pain tend to engage the same processes described by theories of cognitive control in order to solve conceptually similar problems. In the remainder of this Review, we refer to this process as adaptive control, rather than cognitive control, to underscore its broader contribution to negative affect and nociception. In the remainder of this section, we explore the utility of using computationally inspired models of control and reinforcement learning (BOX 2) to clarify the role of aMCC in negative affect and pain. Adapting such models to the study of negative affect and pain promises to enhance the mechanistic specificity of accounts describing this region's putative role in avoidance and defensive behaviours⁵, emotional appraisals²¹, emotional experience¹¹, fear²⁵, attention to pain²⁶, pain expectancy^{126,127}, pain-related motor control^{5,128} and so on.

Control processes are engaged when automatic or habitual responses are insufficient to support goal-directed behaviour¹²⁹. This occurs when there is uncertainty about the optimal course of action (as in situations involving probabilistic learning), when potential actions are associated with substantial risks (for example, of failure, punishment or error) or when there is competition

Event-related potential
(Often abbreviated to ERP.) A scalp-recorded measure of the average brain electrical activity evoked by a particular stimulus or response.

Fear-potentiated startle reflex
A reflex evoked by the sudden onset of high-intensity stimuli (for example, a loud noise) and amplified by negative affect. In humans, this is measured using electrodes overlying orbicularis oculi, the muscle responsible for eye blinks.

Response conflict
Competition elicited by stimuli associated with multiple, incompatible response tendencies — as in the Stroop task.

Box 3 | The role of the aMCC in reward-motivated behaviour and positive affect

The adaptive control hypothesis is broadly consistent with an important body of work detailing the role of anterior midcingulate cortex (aMCC) in reward-motivated behaviour in macaque monkeys. Based on this research, it has been argued that aMCC is critically involved in computing the anticipated reward value of alternative actions, particularly in situations where action–outcome contingencies vary^{35,36}. In particular, there is evidence that neurons in the vicinity of the monkey analogue to the human RCZ are sensitive to errors, the omission of expected rewards and the reward history of alternative responses³⁵.

Computational neurophysiology suggests that these neurons encode predictions about future instrumental rewards, prediction errors in response to discrepancies between expected and obtained rewards and indices of uncertainty that moderate the rate at which new contingencies are learned³⁶.

A key challenge for future research is to determine whether overlap between negative affect, pain and cognitive control in the human aMCC extends to positive affect and reward-motivated behaviour³⁷, as we might expect if this region is insensitive to reinforcer valence and instead computes the salience of both rewards and punishments^{31,73,152,153}. Certainly, the muscles of the upper face do not contribute exclusively to the expression of negative affect²⁰⁸. Moreover, recordings in behaving monkeys suggest that neurons responsive to the anticipation of punishment, reward, or both, are intermingled in aMCC²⁰⁹. Finally, a recent meta-analysis indicates that both reward and punishment consistently activate aMCC in humans²¹⁰.

Future work aimed at surmounting this challenge will require rewards and punishments that are adequately matched and sufficiently potent. A related issue requiring empirical clarification is whether the adaptive control hypothesis pertains equally well to all 'negative' emotions (for example, fear, anger and sadness) (FIG. 3; see [Supplementary information S1](#) (box)).

between plausible alternative actions or between action and inaction (for example, flee-or-freeze, go or no-go). These features are hallmarks of environments where physical threat is genuine, as in many studies of fear, anxiety and pain. Indeed, recent work in rodents demonstrates that physical threat can elicit competition between neural circuits mediating active (Go: avoidance) and passive (No-Go: freezing) defensive behaviours⁵⁷. Not surprisingly, optimal instrumental behaviour in threatening environments has long been thought to require cognitive control^{129,130}, which provides the biasing signal necessary to resolve response uncertainty or competition and avoid potentially catastrophic actions (BOX 2).

On the basis of earlier suggestions^{5,26,29,30,33,34} and more recent computational models of cognitive control and reinforcement learning (BOXES 2,3), we propose that the core function common to negative affect, pain and cognitive control is the need to determine an optimal course of action in the face of uncertainty, that is, to exert control. Based on the data reviewed in the previous section, we further suggest that aMCC implements adaptive control by integrating information about punishment (for example, likelihood and magnitude) arriving from the amygdala, spinothalamic system, striatum, insula and other regions (FIG. 4) in order to bias responding in situations where the optimal course of action is uncertain or entails competition between alternative courses. Outgoing control signals would presumably be sent directly to subcortical and cortical motor centers. Alternatively, control signals generated in aMCC and directed at the amygdala or IPAG might serve to resolve conflict between passive and active defensive behaviours. Several other mechanisms are plausible and these are described more fully in the final section.

The aMCC is responsive to control demands in threatening environments. To date, few studies have addressed whether aMCC is specifically involved in complex action

planning or is sensitive to control demands (for example, the number of response options or response inhibition) in response to perceived physical threat (although somewhat more is known about its role in reward-motivated learning, see BOX 3). Nevertheless, the extant data are consistent with a role in modulating action in response to information about punishment.

First, neuronal recordings in humans and monkeys show that pain-responsive MCC neurons are activated by both anticipation of pain^{131,132} and instrumental escape from pain¹³³. These data underscore the close connections between pain, negative affect elicited by imminent pain and defensive action in MCC.

Second, consistent with the work highlighted in the previous section, other research indicates that these neural signals are sensitive to uncertainty and conflict. For example, source modelling analyses suggest that this MCC activity is amplified by uncertainty about the action associated with pain avoidance (action–outcome uncertainty)¹¹². Likewise, the N2 — an ERP signature of control thought to be generated in MCC — is amplified when pain delivery requires the inhibition of movement¹³⁴ and attenuated when participants are allowed to move^{135,136}.

Third, lesions of the cingulate sulcus in monkeys, which effectively destroy the monkey analogue to the human RCZ, alter how threat modulates ongoing behaviour¹³⁷. Specifically, lesioned monkeys are less reluctant to take food placed above a moving toy snake than controls, an effect reminiscent of that obtained following amygdala lesions^{138,139} and consistent with our suggestion that aMCC exploits ascending punishment signals to modulate instrumental behaviour.

Fourth, recent imaging studies suggest that aMCC might play a more specific part in regulating defensive responses to threat, consistent with our emphasis on control. Across mammalian species, defensive behaviour qualitatively varies with the psychological and physical imminence of threat — distal threats elicit risk assessment, vigilance and the suppression of ongoing appetitive

and consummatory activities (such as foraging). As threat grows more imminent, these behaviours give way to affiliation and alarm calls, flight or freezing (if escape is thwarted) and, ultimately, to active defensive displays or even defensive attack^{77,140–142}. In monkeys, the change of behavioural repertoire in response to increased threat imminence is associated with activation of aMCC¹⁴³. Likewise, in humans, aMCC is activated when escaping from a ‘virtual predator’ whose imminence dynamically varied in a game-like avoidance task (failure to escape the predator was paired with shocks)^{144,145}. These data are consistent with suggestions that aMCC plays a key part in regulating instrumental defensive behaviours³⁴ or is involved in selecting ‘options’ — sequences of elementary actions aimed at accomplishing a goal¹⁴⁶.

Fifth, additional evidence for the adaptive control hypothesis comes from imaging studies showing that aMCC activity during aversively motivated learning is predicted by formal computational models of control and reinforcement learning^{147–150}. Schiller and colleagues¹⁴⁷, for example, recently showed that activation in aMCC encodes punishment prediction errors during the reversal of learned fear. These observations are broadly consistent with the hypothesis that aMCC uses such predictions to adopt the most adaptive response to threat. An alternative possibility is that this effect is a special case of this region’s role in computing signals of reinforcer ‘salience’ during both appetitively and aversively motivated behaviour^{31,151–153} (BOX 3).

A broader perspective on the rostral cingulate cortex. The data that we have reviewed encourage a broader perspective on the functional importance of cingulate activity. The aMCC did not evolve to optimize performance on laboratory measures of ‘cold’ cognition, such as the Stroop task. Indeed, anthropological research suggests that the human brain, like that of our earlier ancestors, evolved in the context of substantial pressure from physical threats, including predation and intraspecific aggression¹⁵⁴, that demanded a neural system capable of flexibly controlling aversively-motivated behaviour. The data we have surveyed are consistent with this speculation and suggest that the contribution of aMCC to laboratory measures of cognitive control might stem from its evolutionarily older role in regulating ‘hot’ behaviours^{25,47,155} — such as expressive behaviour on the face and aversively motivated instrumental learning — that are elicited by stimuli and situations with affective and nociceptive importance (for a related perspective, see REFS 34,156).

This view helps to explain why demands on cognitive control are associated with vestigial defensive reactions, such as brow furrowing and startle, and why individual differences in such measures predict the magnitude of control signals thought to be generated in aMCC (for example, the N2 component of the event-related potential). It also provides an explanation for why the anticipation and receipt of uncertain punishments, which place greater demands on control resources, produce greater activation of aMCC. Regardless of the evolutionary origins, observations such as these are not readily accommodated by accounts that emphasize

a strict segregation of cognition from emotion and nociception in the cingulate.

Limitations of the available evidence

On the basis of a wide range of data and theories, we have suggested that activation of aMCC in studies of negative affect and pain reflects the engagement of control processes that help to optimize responses made in the face of uncertainties about instrumental actions and the outcomes they produce. We have also proposed that aMCC implements adaptive control by synthesizing information about punishment arriving from the amygdala, spinothalamic system, insula and other regions into a biasing signal that could modulate motor centres or subcortical regions, such as the amygdala and IPAG, that more proximally influence active (fleeing) and passive (freezing) defensive behaviours. It is clear that these hypotheses reflect a number of indirect inferences, a limitation that reflects the state of the existing empirical record. Although much work remains, the adaptive control hypothesis provides a clear roadmap to the most profitable avenues for understanding the contribution of aMCC to negative affect and pain. Here, we outline several strategies for more directly testing and refining this account.

First, our meta-analysis demonstrates that aMCC is consistently activated at the subdivision level by manipulations of negative affect, pain and cognitive control. High-resolution, single-subject imaging analyses and intracerebral recordings will be required to determine whether negative affect and pain are anatomically coincident with cognitive control at finer levels of resolution, are intermingled (as some early imaging studies suggest^{157–159}) or are organized along overlapping gradients^{44,160–162}. To permit a more decisive test, such studies should employ a broad battery of well-matched tasks (matched on certainty and motor requirements, for example). The use of single-subject conjunction analyses¹⁶³ or single-subject spatial confidence intervals¹⁶⁴ would provide a rigorous means of quantifying the degree of overlap. Studies of this kind will also prove useful for determining whether negative affect and pain differentially activate superficial (RCZ) versus deep regions of aMCC (BOX 1; see also REF. 137). Based on prior single-subject analyses of negative affect, pain or cognitive control^{126,158,163,165}, we suspect that future work will reveal marked individual differences in the mapping of each domain to cingulate anatomy. Indeed, variation in the location of clusters across individuals within any one domain may well outweigh variation across domains.

Determining the source of such individual differences is a key challenge for future research. One promising way to tackle this problem is to acquire independent measures of affect, pain or cognitive control (for example, eye-tracking, facial action coding or electromyography, pupil dilation and the startle reflex). Variation in such measures — across conditions, across individuals, and within individuals — can clarify the psychological processes that are probabilistically recruited by each domain¹⁶⁶. For example, individuals clearly differ in the intensity or likelihood of negative affect in response to

Prediction error

In reinforcement learning models, an explicit description of the discrepancy between reinforcer expectations and actual reinforcement.

Electromyography

(Often abbreviated to EMG.) Recordings of electrical activity generated by the skeletal musculature.

physical threats⁴⁸ or performance errors⁹³. Although typically not measured, such differences are likely to play a key part in determining which subregions of the cingulate cortex are recruited in each individual. From a translational perspective, such variation may also help to account for differences in treatment response or other clinical features of disorders that are associated with MCC abnormalities, such as post-traumatic stress and bipolar disorders^{10,11}. Already, some investigators have begun to use measures of pain experience (self-report) and expression (peripheral motor reflexes and autonomic activity) to map dimensions of the pain response onto the different subdivisions of the cingulate^{152,167}. Another closely related strategy is to fit computational models to the data acquired from each participant and to use individual differences in the resulting parameter estimates to predict neural activity¹⁶⁸.

Second, complex, multidimensional psychological processes — such as negative affect, pain and cognitive control^{40,169} — are implemented in distributed neural networks. Although aMCC is involved in all three domains, it probably does so in combination with dissociable networks. Functional connectivity¹⁷⁰, mediation¹⁵² or multivoxel pattern¹⁷¹ analyses (MVPA) would permit the identification and dissociation of such networks. MVPA may prove to be a particularly useful tool because it quantifies the degree to which distributed patterns of neural activity encode information about a domain. Using MVPA one can ask, for example, whether the pattern of neural activity corresponding to pain delivery is reactivated by performance errors or threat-of-shock in individual participants. MVPA would also potentially allow the discrimination of domain-specific processes that are intermingled at the subvoxel level¹⁷². Ultimately, such multivariate analyses will be necessary to understand how negative affect, pain and cognitive control emerge from the distributed activity of computationally specialized regions. They should also prove helpful for determining whether the function implemented by aMCC varies qualitatively across different patterns of regional coupling¹⁷⁰.

Some of these regions may reside within the rostral cingulate cortex. We rejected claims of strict functional segregation because our CBMA demonstrated that imaging studies of negative affect, pain and cognitive control consistently activate an overlapping region in aMCC (FIG. 2) and because ACC (the putative 'affective division' of the cingulate) was not preferentially associated with negative affect or pain. Nevertheless, the results of the CBMA are consistent with a measure of functional specialization across rostral cingulate (FIG. 1c). For example, the CBMA indicated that only studies of negative affect consistently activated subgenual ACC (sgACC; see Supplementary information S1 (box)). Likewise, the elicitation of negative affect and pain consistently activated pregenual ACC (pgACC) and posterior MCC (pMCC), whereas cognitive control did not.

Third, thoughtful experimental design, combined with computational modelling and network analyses or more invasive manipulations in non-human animals, will be required to clarify how aMCC uses information

about punishment to adaptively control complex instrumental behaviours. A key question is whether this region represents a monitor, a controller or some combination of the two (BOX 2). It is possible that incoming information about negative affect and pain reflects one of several kinds of inputs that are monitored by aMCC and used to trigger control signals³⁰. Such control signals may be generated in distal regions of the brain, such as the striatum or lateral prefrontal cortex (PFC), or may be generated locally in aMCC and conveyed directly to motor centres. Another possibility is that aMCC directly biases aversively motivated actions through its connections with motor centres, but conveys the need for other kinds of control, such as the biasing of selective attention, to lateral PFC or parietal cortex¹⁷³. Such dissociation would help to reconcile the greater intimacy of aMCC with motor regions while acknowledging the well-documented role of lateral PFC in biasing activity in posterior sensory cortices¹⁷⁴.

A third possibility is that aMCC triggers or implements control in response to insular or amygdalar inputs. Consistent with this, more anxious individuals show aberrant coupling between aMCC and amygdala during the presentation of images known to elicit negative affect¹⁷⁵. Amygdalar signals might reflect competition between passive and active responses to noxious stimuli⁵⁷, punishment predictions or prediction errors¹⁷⁶, or a more general source of information about errors^{177–182}. Such signals could be conveyed directly to aMCC or indirectly, through connections from the amygdala to the striatum, insula or pgACC. Why the amygdala generates such control signals and how this influences, or is influenced by, control processes implemented in aMCC are two crucial questions not addressed by any of the major computational models of control (BOX 2).

Finally, studies of non-human primates and human patients with lesions will be necessary to determine whether the contribution of aMCC to the adaptive control of punishment-motivated instrumental behaviour is a necessary one. Non-human primate research will be particularly useful for bridging the gap between human imaging studies and invasive studies of threat, fear and pain in rabbits and rodents^{25,155} — species that lack certain features of the primate cingulate¹⁸³. Combining invasive techniques with imaging measures in primates should prove particularly useful in this regard. Functional imaging studies would be useful for more precisely identifying functionally homologous regions across species¹⁸⁴. Non-human primate research will also be required to clarify the anatomical connectivity of aMCC, particularly of RCZ, and to develop a more detailed understanding of its role in planning complex actions^{42,185}. This will be particularly crucial owing to the extreme rarity of circumscribed insults to aMCC in humans, a consequence of the wide ramifications of the arterial supply to this region¹³. Although the near absence of such data precludes strong inferences, extant neuropsychological studies are consistent with the idea that MCC makes a necessary contribution to adaptive control in humans¹⁸⁶ (whether this is also true in monkeys remains contentious⁷). In particular, focal damage to left aMCC (including the probable

Neurofeedback

A kind of learning in which real-time neural activity is employed as feedback.

location of RCZ) is associated with exaggerated response conflict and attenuation of the ERN ERP component^{†187,188} (see REF. 186 for conflicting findings). Studies employing neurofeedback techniques¹⁸⁹ or microstimulation to directly manipulate activity in aMCC in humans would be a valuable adjunct to lesion studies. In particular, it would be useful to know whether these more direct manipulations exert similar effects on measures of negative affect, pain and cognitive control.

Conclusions

In summary, a wide variety of evidence demonstrates that negative affect, pain and cognitive control are anatomically and functionally integrated at the subdivision level in aMCC, likely within RCZ, the premotor area harboured in the dorsal portion of aMCC. On this basis, the claim that the cingulate cortex is strictly segregated into cognitive and affective divisions is no longer tenable. Computational models of cognitive control and reinforcement learning provide a foundation for integrating such observations into a mechanistic account of this region's contribution to negative affect and pain.

This framework leads to the adaptive control hypothesis, which suggests that aMCC uses information about punishment to bias responding when the most adaptive course of action is uncertain. This account is not a new theory of rostral cingulate function. Indeed, many of the ideas that we have reviewed are well-known among certain groups of neuroscientists. It is instead a synthesis of earlier suggestions and new data into a clear working hypothesis about the contribution of aMCC to aversively motivated behaviour. As such, we have delineated the kinds of evidence that will be required to refine it.

Perhaps one of the most important challenges is determining whether adaptive control is specific to punishment or, instead, extends to rewards and appetitively motivated behaviour. As we emphasized in BOX 3, a direct test of this possibility using adequately potent, well-matched reinforcers has yet to be performed. To conclude, attempts to refine the adaptive control hypothesis or to adjudicate between it and narrower claims of segregation promise to enrich our understanding of this region's contribution to negative affect and pain in health and disease.

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processes implemented in MCC are likely to be closely related to affective and nociceptive processes.

This paper reviews methods for performing quantitative meta-analyses of functional imaging data using the ALE technique and BrainMap database.

This chapter reviews the anatomy, physiology and function of the cingulate premotor areas, including their connections with the facial nucleus.

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Acknowledgements

We thank the Laboratory for Affective Neuroscience and Waisman Laboratory for Brain Imaging and Behavior staff A. Dinndorf, M. Fox, L. Friedman, L. Hinsenkamp, A. Koppenhaver, A. Laird, B. Nacewicz, D. Rebedew and J.E. Shackman for assistance; M.X. Cohen, W. Irwin, S. Nieuwenhuis, J. Oler, and T. Yarkoni for feedback; and G. Bush for providing details of the meta-analysis described in reference 14. This work was supported by the European Commission (Marie Curie Reintegration Grant to H.A.S.), the University of Toronto Centre for the Study of Pain (Clinician-Scientist award to T.V.S.), Fetzner Foundation (R.J.D.), and National Institute of Mental Health (P50-MH069315, P50-MH084051 and R01-MH43454 (R.J.D.); A.J.S. was partially supported by R01-MH064498 (B.R. Postle); A.S.F. was supported by T32-MH018931 (R.J.D.)).

Competing interests statement

The authors declare no competing financial interests.

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